Chlorpheniramine: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Chlorpheniramine: Patient drug information" and "Chlorpheniramine: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Aller-Chlor [OTC];
- Allergy Relief [OTC];
- Allergy [OTC];
- Chlor-Trimeton Allergy [OTC];
- Chlorphen [OTC];
- Ed Chlorped Jr [OTC];
- FT Allergy Relief [OTC];
- GoodSense Allergy Relief [OTC];
- Pharbechlor [OTC]

Pharmacologic Category

- Alkylamine Derivative;
- Histamine H_1 Antagonist;
- Histamine H_1 Antagonist, First Generation

Dosing: Adult

Motion sickness

Motion sickness (off-label use): Immediate release: **Oral:** 4 to 12 mg administered 3 hours prior to initiating stimulus for motion sickness (Ref). **Note:** Avoid use if it is unsafe for patient to be sedated.

Upper respiratory tract conditions

Upper respiratory tract conditions:

Note: Second-generation H_1-antihistamines are generally preferred due to less sedating and anticholinergic effects. Avoid use in patients with high-risk occupations (eg, pilots, bus drivers) or who may be more prone to anticholinergic effects (eg, older adults) (Ref).

Immediate release: **Oral:** 4 mg every 4 to 6 hours; do not exceed 24 mg/24 hours.

Extended release: **Oral:** 12 mg every 12 hours; do not exceed 24 mg/24 hours.

Urticaria, new onset and chronic spontaneous

Urticaria, new onset and chronic spontaneous (alternative agent) (off-label use):

Note: Second-generation H_1-antihistamines are preferred due to less sedating and anticholinergic effects (Ref). May consider use in combination with a second-generation H_1-antihistamine in patients in whom bedtime sedating effects may be beneficial; avoid use in patients with high-risk occupations (eg, pilots, bus drivers) or who may be more prone to anticholinergic effects (eg, older adults) (Ref). Dosing is based on manufacturer's labeling for use in upper respiratory tract conditions.

Immediate release: **Oral:** 4 mg every 4 to 6 hours; do not exceed 24 mg/24 hours.

Extended release: **Oral:** 12 mg every 12 hours; do not exceed 24 mg/24 hours.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: Limited information exists regarding the disposition of chlorpheniramine in patients with kidney impairment.

Altered kidney function:

eGFR >30 mL/minute/1.73 m^2: No dosage adjustment necessary (Ref).

eGFR ≤30 mL/minute/1.73 m^2: No initial dosage adjustment necessary; however, greatly prolonged half-lives in patients with severe kidney disease have been reported; use with caution (Ref).

Hemodialysis, intermittent (thrice weekly): Unlikely to be dialyzed (large V_d): Dose as for eGFR ≤30 mL/minute/1.73 m^2 (Ref).

Peritoneal dialysis: Unlikely to be dialyzed (large V_d): Dose as for eGFR ≤30 mL/minute/1.73 m^2 (Ref).

CRRT: Dose as for eGFR ≤30 mL/minute/1.73 m^2 (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Dose as for eGFR ≤30 mL/minute/1.73 m^2 (Ref).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling; however, chlorpheniramine is metabolized via the liver; therefore, a dose adjustment may be necessary.

Dosing: Older Adult

Avoid use (Ref).

Dosing: Pediatric

(For additional information see "Chlorpheniramine: Pediatric drug information")

Dosage guidance:

Safety: Safety and efficacy for the use of cough and cold products in infants and young children are limited; the AAP warns against the use of these products for respiratory illnesses in infants and young children; the FDA does not recommend OTC use in infants and children <2 years of age due to the risk of serious and life-threatening adverse effects (including death) and recommends to use with caution in pediatric patients ≥2 years of age (Ref).

Upper respiratory allergy symptoms [hay fever]

Upper respiratory allergy symptoms [hay fever]:

Immediate release:

Oral syrup (eg, Ed Chlorped Jr): 2 mg chlorpheniramine per 5 mL:

Children 2 to <6 years: Limited data available: Oral: 1 mg every 4 to 6 hours; maximum daily dose: 6 mg/**day** (Ref).

Children 6 to <12 years: Oral: 2 mg every 4 to 6 hours; maximum daily dose: 12 mg/**day**.

Children ≥12 years and Adolescents: Oral: 4 mg every 4 to 6 hours; maximum daily dose: 24 mg/**day**.

Tablets (eg, Allergy, Allergy Relief, Chlor-Trimeton, GoodSense Allergy):

Children 6 to <12 years: Oral: 2 mg every 4 to 6 hours; maximum daily dose: 12 mg/**day**.

Children ≥12 years and Adolescents: Oral: 4 mg every 4 to 6 hours; maximum daily dose: 24 mg/**day**.

Extended release: Tablet (eg, Chlor-Trimeton Allergy): Children ≥12 years and Adolescents: Oral: 12 mg every 12 hours; maximum dose: 24 mg in 24 hours.

Dosing: Kidney Impairment: Pediatric

Children ≥6 years and Adolescents: Oral: There are no pediatric dosage adjustments provided in manufacturer's labeling; in adult patients with severe kidney disease, greatly prolonged half-lives have been reported; use with caution (Ref).

Dosing: Liver Impairment: Pediatric

Children ≥6 years and Adolescents: Oral: There are no dosage adjustments provided in manufacturer's labeling; however, chlorpheniramine is metabolized via the liver; therefore, a dose adjustment may be necessary.

Adverse Reactions

There are no adverse reactions listed in the manufacturer's labeling.

Contraindications

OTC labeling: When used for self-medication, do not use to make a child sleep.

Warnings/Precautions

Concerns related to adverse effects:

• CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving).

Special populations:

• Bariatric surgery: Presurgical assessment of the indication for use, symptoms, and goals of therapy should be documented to enable postsurgical assessment. Monitor for continued efficacy after bariatric surgery and consider switching to an alternative formulation or medication if symptoms worsen.

• Pediatric: Antihistamines may cause excitation in young children.

Dosage form specific issues:

• Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.

• Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated with hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Zar 2007).

Other warnings/precautions:

• Self-medication (OTC use): When used for self-medication (OTC), notify health care provider prior to use if you have breathing problems (eg, chronic bronchitis, emphysema), glaucoma, difficultly in urination due to enlargement of the prostate gland, or are currently taking sedatives, tranquilizers.

Warnings: Additional Pediatric Considerations

Safety and efficacy for the use of cough and cold products in pediatric patients <4 years of age is limited; the AAP warns against the use of these products for respiratory illnesses in young children. Serious adverse effects including death have been reported. Many of these products contain multiple active ingredients, increasing the risk of accidental overdose when used with other products. The FDA does not recommend OTC uses for these products in pediatric patients <2 years of age and recommends to use with caution in pediatric patients ≥2 years of age. Health care providers are reminded to ask caregivers about the use of OTC cough and cold products in order to avoid exposure to multiple medications containing the same ingredient (AAP 2018; CDC 2007; FDA 2017; FDA 2018).

Some dosage forms may contain propylene glycol; in neonates large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Syrup, Oral, as maleate:

Ed Chlorped Jr: 2 mg/5 mL (473 mL) [alcohol free, sugar free; contains fd&c red #40 (allura red ac dye), methylparaben (methyl p-hydroxybenzoate), propylene glycol, propylparaben (propyl p-hydroxybenzoate); cherry flavor]

Tablet, Oral, as maleate:

Aller-Chlor: 4 mg [scored; contains quinoline (d&c yellow #10) aluminum lake]

Allergy: 4 mg [scored; contains corn starch, quinoline (d&c yellow #10) aluminum lake]

Allergy Relief: 4 mg [contains corn starch, quinoline (d&c yellow #10) aluminum lake]

Allergy Relief: 4 mg [scored; contains corn starch, quinoline (d&c yellow #10) aluminum lake]

Chlorphen: 4 mg [scored; contains corn starch, quinoline (d&c yellow #10) aluminum lake]

FT Allergy Relief: 4 mg [contains corn starch, quinoline (d&c yellow #10) aluminum lake]

GoodSense Allergy Relief: 4 mg [contains quinoline (d&c yellow #10) aluminum lake]

Pharbechlor: 4 mg

Generic: 4 mg

Tablet Extended Release, Oral, as maleate:

Chlor-Trimeton Allergy: 12 mg [contains fd&c blue #2 (indigo carm) aluminum lake, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Generic: 12 mg

Generic Equivalent Available: US

Yes

Pricing: US

Tablet, controlled release (Chlor-Trimeton Allergy Oral)

12 mg (per each): $0.46

Tablet, controlled release (Chlorpheniramine Maleate ER Oral)

12 mg (per each): $0.63

Tablets (Chlorpheniramine Maleate Oral)

4 mg (per each): $0.01

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Administration: Adult

Oral:

May be administered with food or water. Timed release oral forms are to be swallowed whole, not crushed or chewed. Administer liquid formulations with an accurate measuring device; do not use a household teaspoon (overdosage may occur). If used for motion sickness, administer 3 hours before stimulus.

Bariatric surgery: Chlorpheniramine is available as an ER formulation. Bariatric surgery may significantly alter the release characteristics in an unknown manner. Chlorpheniramine is also available in an IR formulation. Oral solutions may contain nonabsorbable sugars (eg, sorbitol, mannitol, xylitol) that can cause dumping syndrome (Ref). Refer to product labeling. Providers should determine if the condition being treated can be safely monitored or if a switch to an alternative formulation or medication is necessary (Ref).

Administration: Pediatric

Oral:

Immediate release:

Tablet: May be administered with food or water.

Syrup: Administer liquid formulations with an accurate measuring device; do not use a household teaspoon (overdosage may occur).

Extended release: **Tablet**: Swallow whole; do not crush or chew. May be administered with food or water.

Storage/Stability

Store at room temperature.

Use: Labeled Indications

Upper respiratory tract conditions: Temporary relief of symptoms (runny nose; sneezing; itching of the eyes, nose, or throat) associated with the hay fever (allergic rhinitis) and other upper respiratory tract conditions.

Use: Off-Label: Adult

Motion sickness; Urticaria, new onset and chronic spontaneous

Medication Safety Issues

Sound-alike/look-alike issues:

Chlor-Trimeton may be confused with Chloromycetin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs (pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

Older Adult: High-Risk Medication:

Beers Criteria: Chlorpheniramine, a first-generation antihistamine, is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) because of its potent anticholinergic properties resulting in increased risk of confusion, dry mouth, constipation, and other anticholinergic effects or toxicity; use should also be avoided because of reduced clearance with advanced age and tolerance associated with use as a hypnotic. Exposure to concurrent anticholinergic drugs also increases risk of falls, delirium, and dementia; consider total anticholinergic burden when conducting medication reviews (Beers Criteria [AGS 2023]).

Metabolism/Transport Effects

Substrate of CYP2D6 (Major), CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acetylcholinesterase Inhibitors: May decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Aclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Acrivastine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ajmaline: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amantadine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Amezinium: Antihistamines may increase stimulatory effects of Amezinium. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Artemether and Lumefantrine: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Benperidol. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benztropine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Benztropine. *Risk C: Monitor*

Benzylpenicilloyl Polylysine: Coadministration of Antihistamines and Benzylpenicilloyl Polylysine may alter diagnostic results. Management: Suspend systemic H1 antagonists for benzylpenicilloyl-polylysine skin testing and delay testing until systemic antihistaminic effects have dissipated. A histamine skin test may be used to assess persistent antihistaminic effects. *Risk D: Consider Therapy Modification*

Betahistine: Antihistamines may decrease therapeutic effects of Betahistine. Betahistine may decrease therapeutic effects of Antihistamines. *Risk C: Monitor*

Biperiden: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Biperiden. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Bornaprine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bromperidol. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buclizine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Buclizine. *Risk C: Monitor*

Buclizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Certoparin: Antihistamines may increase therapeutic effects of Certoparin. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Chlorprothixene. *Risk C: Monitor*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cimetropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Cimetropium. *Risk X: Avoid*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of CloZAPine. Management: Consider alternatives to this combination whenever possible. If combined, monitor closely for signs and symptoms of gastrointestinal hypomotility and consider prophylactic laxative treatment. *Risk D: Consider Therapy Modification*

CNS Depressants: Chlorpheniramine may increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclizine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

CYP2D6 Inhibitors (Strong): May increase serum concentrations of Chlorpheniramine. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Darifenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Darifenacin. *Risk C: Monitor*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

DexmedeTOMIDine: CNS Depressants may increase CNS depressant effects of DexmedeTOMIDine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Dicyclomine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dicyclomine. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimethindene (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroNABinol: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of DroNABinol. *Risk X: Avoid*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Eluxadoline: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Eluxadoline. *Risk X: Avoid*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Fesoterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Fesoterodine. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

FluPHENAZine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Fosphenytoin-Phenytoin: Chlorpheniramine may increase serum concentrations of Fosphenytoin-Phenytoin. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

Gastrointestinal Agents (Prokinetic): Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepotidacin: May decrease anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Glucagon: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Glucagon. Specifically, the risk of gastrointestinal adverse effects may be increased. *Risk C: Monitor*

Glycopyrrolate (Oral Inhalation): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Oral Inhalation). *Risk X: Avoid*

Glycopyrrolate (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Systemic). *Risk C: Monitor*

Glycopyrronium (Topical): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase adverse/toxic effects of Agents with Clinically Relevant Anticholinergic Effects. Specifically, intraocular pressure may be increased. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Haloperidol: QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of Haloperidol. *Risk C: Monitor*

Hyaluronidase: Antihistamines may decrease therapeutic effects of Hyaluronidase. *Risk C: Monitor*

HydrOXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hyoscyamine: Anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects and Hyoscyamine may increase with coadministration. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Ipratropium (Nasal): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ipratropium (Oral Inhalation): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Isoproterenol: Chlorpheniramine may increase therapeutic effects of Isoproterenol. *Risk C: Monitor*

Itopride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Itopride. *Risk C: Monitor*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levosulpiride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Levosulpiride. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mavorixafor: May increase serum concentrations of CYP2D6 Substrates (High risk with Inhibitors). *Risk X: Avoid*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Melperone: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methscopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methscopolamine. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirabegron: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Mirabegron. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase adverse/toxic effects of Antihistamines, First Generation. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nitroglycerin: Agents with Clinically Relevant Anticholinergic Effects may decrease absorption of Nitroglycerin. Specifically, anticholinergic agents may decrease the dissolution of sublingual nitroglycerin tablets, possibly impairing or slowing nitroglycerin absorption. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANZapine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OLANZapine. *Risk C: Monitor*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

OxyBUTYnin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OxyBUTYnin. *Risk C: Monitor*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Perphenazine. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pitolisant: Antihistamines may decrease therapeutic effects of Pitolisant. *Risk X: Avoid*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Potassium Chloride: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Chloride. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium chloride. *Risk X: Avoid*

Potassium Citrate: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Citrate. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium citrate. *Risk X: Avoid*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pramlintide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. These effects are specific to the GI tract. *Risk X: Avoid*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propantheline: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Propantheline. *Risk C: Monitor*

Propiverine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QUEtiapine: CNS Depressants may increase CNS depressant effects of QUEtiapine. *Risk C: Monitor*

QuiNIDine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Ramosetron: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Ramosetron. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Revefenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Revefenacin. *Risk X: Avoid*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Rivastigmine: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Rivastigmine. Rivastigmine may decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Management: Use of rivastigmine with an anticholinergic agent is not recommended unless clinically necessary. If the combination is necessary, monitor for reduced anticholinergic effects. *Risk D: Consider Therapy Modification*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Secretin: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Secretin. Management: Avoid concomitant use of anticholinergic agents and secretin. Discontinue anticholinergic agents at least 5 half-lives prior to administration of secretin. *Risk D: Consider Therapy Modification*

Sofpironium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Sofpironium. *Risk X: Avoid*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Agents with Clinically Relevant Anticholinergic Effects may increase serum concentrations of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Thiothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Thiothixene. *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

Tiapride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Tiapride. *Risk C: Monitor*

Tiotropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tiotropium. *Risk X: Avoid*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Tolterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tolterodine. *Risk C: Monitor*

Topiramate: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Topiramate. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

Tranylcypromine: May increase anticholinergic effects of Antihistamines, First Generation. *Risk X: Avoid*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Trospium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trospium. *Risk C: Monitor*

Umeclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Outcome data following maternal use of chlorpheniramine during pregnancy are available (Aselton 1985; Diav-Citrin 2003; Gilboa 2009; Gilboa 2014; Heinonen 1977; Jick 1981; Li 2013).

Algorithms are available for the treatment of acute rhinitis and urticaria. First-generation oral antihistamines are generally not recommended for use in pregnant patients due to side effects (AAAAI/ACAAI [Dykewicz 2020]; EAACI [Zuberbier 2022]). However, chlorpheniramine may be used when a first-generation antihistamine is needed during pregnancy (Murase 2014; Stefaniak 2022).

Breastfeeding Considerations

It is not known if chlorpheniramine is present in breast milk.

Drowsiness and irritability have been reported in breastfed infants exposed to antihistamines (Ito 1993). In general, if a breastfed infant is exposed to a first-generation antihistamine via breast milk, they should be monitored for irritability or drowsiness (Butler 2014).

Antihistamines may temporarily decrease maternal serum prolactin concentrations when administered prior to the establishment of breastfeeding (Messinis 1985).

Use of a second-generation antihistamine is preferred when an oral antihistamine is needed in lactating patients (Butler 2014; EAACI [Zuberbier 2022]).

Mechanism of Action

Competes with histamine for H_1-receptor sites on effector cells in the gastrointestinal tract, blood vessels, and respiratory tract

Pharmacokinetics (Adult Data Unless Noted)

Note: Data from chlorpheniramine maleate:

Distribution: V_d: Children and Adolescents 6 to 16 years: 7 ± 2.8 L/kg (Simons 1982); Adults: 6-12 L/kg (Paton 1985)

Protein binding: ~70% (Rumore 1984).

Metabolism: Hepatic via CYP450 enzymes (including CYP2D6 and other unidentified enzymes) to active and inactive metabolites; significant first-pass effect (Sharma 2003)

Half-life elimination: Serum: Children and Adolescents 6 to 16 years: 13.1 ± 6.6 hours (range: 6.3 to 23.1 hours) (Simons, 1982); Adults: 14-24 hours (Paton 1985)

Time to peak: Children and Adolescents 6 to 16 years: Oral : 2.5 ± 1.5 hours (range: 1 to 6 hours) (Simons 1982); Adults: 2-3 hours (Sharma 2003)

Excretion: Urine (Sharma 2003)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Severe chronic kidney impairment: Half-life prolonged to 280 to 330 hours (Paton 1985).

Patient Counseling Points

(For additional information see ”Chlorpheniramine: Patient drug information”)

What is this drug used for?

• It is used to ease allergy signs.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Fatigue

• Anxiety

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Allerfin | Anallerge | Chlorohistol | Histan | Histop | Omvil | Pirafene;
- (AR) Argentina: Alergitrat | Desenfriolitos;
- (BD) Bangladesh: Adhista | Alejess | Alergyl | Alkamin | Allerfin | Allergex | Allergo | Allermine | Anti-Hist | Antismine | Antista | Biocin | Clomin | Dayhista | Denamine | Distamin | Dolcin | Eramin | Ethista | Expilin | G-antihistamin | Hisnol | Hispacin | Histacin | Histaco | Histadex | Histal | Histalex | Histalock | Histamex | Histamin | Histamol | Histaseem | Histex | Histocin | Histolet | Histrol | Hitagen | Lergigan | Mystacin | Niramin | Pheramin | Piriton | Promalin | Remacin | Sedilux m | Servinin | Tymin | Winkol;
- (BE) Belgium: Kelargine;
- (BF) Burkina Faso: Chlorphenamine;
- (BZ) Belize: Chlorpheniramine | Clorfeniramina fd;
- (CL) Chile: Clorfenamina maleato | Clorferaton | Clorprimeton | Histafed | Nipolen | Prodel | Respide | Scadan;
- (CN) China: Chlor trimeton | Chlorpheniramine c | Chlorpheniramine maleate | Mei min | Yi ke an;
- (CO) Colombia: Anti allergy | Clodryl | Clorfam | Clorfeniramina | Clorfeniramina maleato | Clorotrimeton | Codilan | Hismedex | Isticol;
- (CR) Costa Rica: Clorfenamina | Clorfenamina maleato | Clorfeniramina | Clorfeniramina chemo | Clorfeniramina maleato | Clorfeniramina maleato cslox | Clorfeniramina maleato fb | Clorfeniramina raven | Histaprin s/c;
- (DO) Dominican Republic: Clorfeniramina | Clorotrimeton;
- (EC) Ecuador: Clorfenamina | Histal | Kronodryl | Nipolen;
- (EE) Estonia: Alergamons | Chlorpheniramine | Piriton;
- (EG) Egypt: Allergyl | Anallerge | Chlorantine m | Chlorpheniramine | Pirafene;
- (ET) Ethiopia: Aspi m | Chlorphenamine | Chlorpheniramine maleate | Cipium;
- (GB) United Kingdom: Allercalm | Allerief | Alunex | Boots allergy relief | Boots allergy relief 1 year plus | Boots children's allergy relief antihistamine | Calimal | Chlorphenamine | Chlorphenira almus | Chlorpheniram | Chlorpheniram cox | Chlorpheniram dc | Chlorpheniram kent | Hayleve | Piriton | Pollenase | Rhino syrup allergy;
- (GR) Greece: Istamex;
- (GT) Guatemala: Clorfeniramina maleato | Histaprin s/c;
- (HK) Hong Kong: Allergex | Allermin | Allersin f | Antamin | Antihist | Antimine | Antitamin | Apomin | Aromin | Axcel chlorpheniramine | Bf-Chlorphenamine | Chlor-Diamine | Chloramine | Chlormale | Chlormine | Chlorminol | Chloroton | Chlorpheniramine | Chlorpheniramine maleate | Chlorpyrimine | Coriton | Euromine | Flentist | Histal | Horamine | Marpheniramine | Piriton | Qualiton | Sprinsol | Telamin | U b | U-Chlormine | Uni-Ramine | Vida-His;
- (ID) Indonesia: Alermax | Alleron | Chlorpheniramine | Chlorphenon | Cohistan | Ctm | Dehista | Hufaphenon | Klorfeniramina | Kofiren | Metachlor | Orphen | Paraphenon | Pehachlor | Piriton | Zecamex;

- (IE) Ireland: Anti Hist | Piriton;
- (IL) Israel: Ahiston;
- (IN) India: Allercalm | Allerkil | Alphinine | Cadistin | Cepiam-tr | Chlorfen | Chlorpheniramine | Chlorpheniramine maleate | Coldnil | Cpmine | Histanil;
- (IQ) Iraq: Alerdain | Allerjix | Antihist | Chlomal | Cloramin mdi | Histadin | Histofen | Klofenamin awa | Poramine | Safastin;
- (IT) Italy: Trimeton | Trimeton AR;
- (JO) Jordan: Allerfin | Anallerge | Chlorhistol | Histanore | Istamex | Jofamin;
- (JP) Japan: Allergin | Bismilla | C meton | Chlodamin | Chlometon | Chlor trimeton | Chloramine hexal | Chlorphenamine Maleate Daiko | Chlorphenamine maleate merck hoei | Chlorphenamine Maleate Nipro | Cibena | Histal | Kohiston | Malemirane r hexal | Neo restar fuji | Neorestamin | Reston | Teiporin | Telgin;
- (KE) Kenya: Allercol | Allerex | Allergex | Allerief | Alleston | Anhistina | Aspi m | Cariton | Chaleate | Chlorhist | Chlorpheniramine | Chlorpheniramine maleate | Cipi m | Cipium | Dawa cpm | Dinlamin | Phenalin | Phenamine | Piriclor | Pirioss | Piriton | Rinalin | Rinovil | Seepem | Toramin | Unihist;
- (KR) Korea, Republic of: Chloramine | Peniramin | Pheniramine;
- (KW) Kuwait: Allergetin | Allergyl | Anallerge | Chlorohistol | Histan | Histop | Omvil | Pheniram;
- (MX) Mexico: Alerdil | Antadex h | Blendox | Clorfenamin ultra | Clorfenamina | Clorfenamina gi br | Cloro trimeton | Cronal | Derimeton | Docsi | Padamina | Puntura;
- (MY) Malaysia: Aller | Allermine | Allerphen | Allersin | Allersin f | Alleryl | Antamin | Antismine | Axcel Chlorphenira | Axcel chlorpheniramine | Chloriton | Chlorpheniramine | Chlorpheniramine maleate | Chlorpyrimine | Cloramina | Dyriton | Hismine | Horamine | Kotamin | Piri Expectorant | Piriton | Sensitamin | Tromine | Zoramine;
- (NG) Nigeria: Ac fed | Actnaza | Allergin | Areactin | Avro chlorpheniramine maleate | Bond cpm | Chlorpheniramine maleate | Dana cpm | Fenclor syrup | Gauze chlorpheniramine maleate | Labriton | Mobitin | Mopson chlorpheniramine | Prytune | Sam chlorpheniramine | Shree mansifed | Sinufed cold paediatric | Stactifed | Sterlin chlorpheniramine maleate | Supritin | Vamafed | Zunafed;
- (NZ) New Zealand: Histafen;
- (OM) Oman: Omvil;
- (PA) Panama: Aler g | Clorfen | Clorfeniramina | Clorfeniramina mp | Diclorfen | Fenaler | Histaprin s/c | Nipolen | Prilon;
- (PE) Peru: Allermed | Antihistamin | Bioalergan | Biolergan | Ciralerg | Clorfenamina | Clorfenamina maleato | Clorfenamina maleato fmndtria | Clorhistamin | Cloro alergan | Cloro toxin | Cloroalerfin | Clorotrimeton | Clorph allergy | Clorph max | Corifan | Histaclorf | Histafed | Histaton | Lergimax | Namine | Nastimed | Scadan;
- (PH) Philippines: Allermax | Antamin | Barominic | Chlor trimeton | Chlorphenamimine maleate | Chlorphenamine | Chlorpheniramine | Clormetamine | Cohistan | Corantish | Fahrenal | Histapen | Mervilar | Riphen | Synestal | Ul chlorphenamine | Valemine;
- (PK) Pakistan: Afdaphen | Alerphene | Allergex | Allergine | Allergon | Allermin | Allerphene | Apiton | Bariton | Briphen | Cfr | Chlorpheniramin | Chlorpheniramine maleate | Chlortab | Cpm | Fenamin | Fenram | Histagic | Histalif | Histalon | Histamol | Histanil | Multivil | Oceton | Panagic | Phemin | Phenamine | Phenermin | Piriton | Ploraton | Rest | Unihist;
- (PR) Puerto Rico: Allergy | Allergy Relief | Allergy Time | Chlor trimeton | Chlorpheniramine | Chlorpheniramine maleate | Ed chlorped pediatric | Ed Chlortan | Pharbechlor | Teldrin hbp;
- (PY) Paraguay: Analer | Biocort | Clorfen | Clorfeniramina dasanti | Clorfeniramina dutriec | Clorfeniramina heisecke | Clorfeniramina maleato mintlab | Clorfeniramina maleato quimfa | Clorhistam | Flurox | Nestaler | Romaler;
- (QA) Qatar: Allerfin | Chlorohistol | Histan | Omvil | Piriton;
- (RO) Romania: Clorfeniramin;
- (SA) Saudi Arabia: Allerfin | Anallerge | Chlorohistol | Chlorpheniramine maleate | Histan | Histop | Lergicol | Pheniram | Pirafene | Zistan;
- (SG) Singapore: Allermin | Allerphen | Allersin | Allersin f | Alleryl | Antamin | Axcel chlorpheniramine | Chloramine | Chlormine | Chlorpheniramine | Chlorpheniramine maleate | Chlorpyrimine | Clomine | Horamine | Litacidin | Piriton;
- (SR) Suriname: Chlorphenamine | Chlorpheniramine | Chlorpheniramine maleate | Histal;

- (SV) El Salvador: Clorfeniramina | Clorfeniramina Gamma | Fenaler;
- (TH) Thailand: Allergin | Aly | Antamin | Anteehist | Antihist | Arc-chlor | Bevon | Cafen | Chlophe | Chlophen | Chlor | Chlorahist | Chloramin | Chloramine S | Chlordon | Chlorhist | Chlorimin | Chloriton | Chlorleate | Chlormin | Chlormine | Chlorphemine | Chlorphen | Chlorphen-y | Chlorphenamine | Chlorpheniramine | Chlorpheno | Chlorpheril | Chlorpyrimine | Chlortab g | Chlortab gr | Chlortab gray | Chlortab r | Chlortrimin | Chlovelon | Cohistan | Collumine | Coriton | Diabemed | Hisdaron | Histatab | Ilcid | Iwatt | Jcl chlorphen | Kloramin | Kresschlorphen | Lichlorphen | Manohista | Medcor | P.k. ramin | Patarphen | Phedamin | Pheramine | Piriton | Pophen | Pyramine | Ramine | Sichlorphen | Sinchlormin | Suramine;
- (TN) Tunisia: Chlorohistol;
- (TR) Turkey: Alerfin;
- (TW) Taiwan: Allermin | Anti-amine | Arellmin | Beina | Benda | Chlopamine | Chloramine | Chlorimin | Chlorpheniramine | Chlorpheniramine maleate | Com-trimeton | Demin | Dexferin | Evenin | Heamin | Lontomin | Malenate | Neo antihistamine | Neo benamin | Niramine | Phenamin | Pheniramine | Ronamin;
- (UA) Ukraine: Suprostilin;
- (UG) Uganda: Ago cpm | Allergex | Aspi m | Chaleate | Chlorpheniramine | Cipium | Cough linctus | Dawa cpm | Koflin | Phenirex | Piriton | Rinalin;
- (UY) Uruguay: Alercel | Analerg | Argisan | Clorfeniramina | Clorfeniramina Callion & Hamonet | Clorfeniramina dorrego | Clorfeniramina maleato farmaco uruguayo | Clorotrimeton | Hibernyl | Kalitron | Solucion de clorfeniramina dorrego;
- (VE) Venezuela, Bolivarian Republic of: Clorfeniramina | Clorotrimeton | Clorotrimeton inyectable | Diclorfam | Inquiramin;
- (VN) Viet Nam: Agitec f | Allerfar;
- (ZA) South Africa: Allergex | Chlorhist | Chlorphen | Rhineton;
- (ZM) Zambia: Cadiphen | Chlorpheniramine | Cipium | Dawa cpm | Horamine | Metadil | Milmaleate | Rinalin;
- (ZW) Zimbabwe: Allergex | Chlorpheniramine | Sensitamine

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